



COMP 003 – COMPUTER PROGRAMMING 2
SECOND SEMESTER, AY 2025-2026

FINAL LABORATORY ACTIVITY NO. 1

Name: _____ **Score:** _____

Name: _____ **Program:** _____

Faculty Asst. Prof. Ma. Ymelda Batalla **Course Code - Section:** _____

Final Laboratory Activity Name: **Daily Temperature Monitoring and Analysis System Using C++ Vector**

Learning Outcomes:

- Upon completion of the activity, the students should be able to:
- Demonstrate the ability to apply theoretical concepts learned in vector in C++ programming language through an accurate program source code;
 - Analyze the program results critically and interpret the data to validate accurate output; and
 - Collaborate and cooperate in a team to accomplish the group task.

Instructions:

- This activity is composed of two (2) students per group.
- Create a C++ program that will provide the following parameters to the users:
 - Prompt the user to enter the number of temperature readings captured for a specific day;
 - Validate all the user inputs:
 - Temperature readings must be greater than zero
 - Temperature readings must be numeric and within a realistic range
 - Store all temperature readings using a C++ Vector;
 - Display all recorded temperature;
 - Compute and display the following:
 - Average temperature per day;
 - Highest temperature;
 - Lowest temperature;
 - Number of temperature readings beyond 40 degrees Celsius;
 - Sort temperature in ascending order;
 - Categorize temperatures:
 - Cold
 - Hot
 - Normal
 - Display necessary output
- Allow the user to repeat the process for another day until the user selects to stop the program
- Note: Utilize vector, UDF, and do not use algorithm header file

The following criteria will be used to check the group’s work or output:

CRITERIA	EXCELLENT (3)	FAIR (2)	POOR (1)	SCORE
Functionality (x5)	The program works and meets all the required requirements.	The program works and partially meets the specified requirements.	The program does not meet all the required requirements.	
Output (x3)	The program displays accurate and all the required output.	The program displays minimal output.	The program does not display any output.	
Readability (x1)	The program code is well organized and can easily be understood.	The program code is somewhat organized and understood.	The program code is poorly written.	
Documentation (x1)	The program code is well commented.	The program code lacks the major comments that make the code difficult to understand.	The program code does not have any proper documentation.	